OVERHEAD CRANE INSTALLATION SERVICE

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INSTALLATION

Overhead Cranes

Overhead crane installation is a multi-stage process that begins with a thorough site assessment and concludes with a mandatory load test at 110% of the crane's rated capacity. The procedure typically takes between 2 and 10 days, though complex systems can require up to a month.

Phase 1: Site Assessment and Preparation

Before any hardware arrives, installers and customers must coordinate on facility readiness:

- **Site Inspection:** Technicians verify foundation stability, measure the runway span against approval drawings, and identify any obstructions like ductwork, piping, or machinery.
- **Runway Verification:** Existing runway beams are inspected for alignment, as misalignment can cause premature component wear.
- **Logistics Planning:** Customers must clear work areas, ensure doors are accessible, and prepare staging areas for equipment and components.
- **Utility Confirmation:** The facility must have the correct power source and specifications ready before the scheduled installation date.

Phase 2: Assembly and Hoisting

The physical installation involves assembling massive components and lifting them into place:

- **Base Assembly:** Bridge beams and end trucks are cleaned, inspected, and assembled on the ground.
- **Hoisting:** Using mobile cranes, existing overhead cranes, or rigging equipment, the main bridge beam is lifted and positioned onto the runway rails.
- **Component Integration:** The electric hoist is attached to the bridge beam, and drive motors are installed, often requiring lubrication of gearboxes and placement of shockproof pads.
- **Electrical Wiring:** Technicians connect the power supply to hoist and trolley motors, install control boxes, and configure control panels. Modern systems may use "aviation plugs" to simplify wiring and maintenance.

Phase 3: Testing and Commissioning

Once the crane is physically installed, it must undergo rigorous safety and performance checks:

- **Electrical Safety Test:** All connection points are checked and tightened before the power is switched on to adjust the electrical line.
- **Range of Motion:** The crane is run through its full range of motion to verify clearances and identify unusual noises or malfunctions.
- **Load Testing:** A critical final step is a **110% load test** to verify the crane can statically hold weight without brake slippage.
- **Safety Verification:** Technicians confirm the functionality of limit switches, emergency stops, and warning devices.

Common Installation Obstacles

Installers frequently address facility-specific challenges during the process:

- **Headroom Issues:** Adjusting the crane or hoist to fit within limited vertical space.
- **Accessibility:** Maneuvering around immovable utilities such as conveyor belts, pits, or piping.
- **Equipment Use:** Coordinating the use of onsite forklifts, scissor lifts, or welders to assist the installation team.